

Model selection

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Warmup

Work on the warmup activity (handout).

Warmup

- Deviance on training data is biased estimate of performance on new data
- Deviance on training data can never increase when I add additional variables

Which model would you prefer?

```
m_dengue1 <- glm(Dengue ~ Age + WBC + PLT,  
                  data = dengue,  
                  family = binomial)  
m_dengue1$deviance
```

← smaller AIC

```
## [1] 5200.823
```

$$AIC = 5200.823 + 2(4) = 5208.823$$

```
m_dengue2 <- glm(Dengue ~ Age + WBC + PLT + HCT,  
                  data = dengue, family = binomial)  
m_dengue2$deviance
```

```
## [1] 5199.946
```

$$AIC = 5199.946 + 2(5) = 5209.946$$

Another example

```
m_dengue1 <- glm(Dengue ~ Age + WBC + PLT,  
                  data = dengue,  
                  family = binomial)
```

```
m_dengue1$aic
```

```
## [1] 5208.823
```

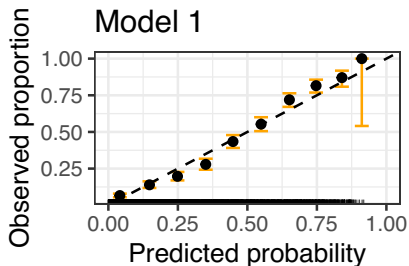
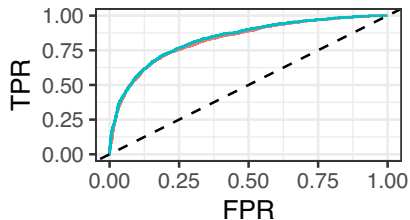
```
m_dengue2 <- glm(Dengue ~ Age + WBC + PLT +  
                  BMI + Temperature + Vomiting,  
                  data = dengue, family = binomial)
```

```
m_dengue2$aic
```

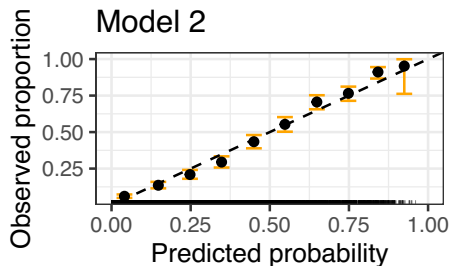
```
## [1] 5107.318
```

Another example

ROC & calibration plots for predictions from 10-fold CV



— Model 1 — Model 2



A difference in AIC
≠
a clinically meaningful
difference in Dengue
prediction

Key points

- ▶ Deviance (on the training data) can **never** increase when I add additional explanatory variables – don't use raw deviance to compare models with different numbers of predictors
- ▶ AIC is a better method to compare models (estimate of predictive ability on new data)
- ▶ But, a difference in AIC does not always have **practical significance**
- ▶ Other criteria for choosing a model: type of research question, model interpretability, model parsimony, ease of use by practitioners, model diagnostics

Types of research question

For what different types of research question might I use a regression model?

Prediction: Given X , can we predict Y ?

- Interpretability is less important
- Assumptions are less important
- Predictive ability is more important
- Parsimony may be less important

Causal inference: If I change X , how does Y change?

- requires very strong assumptions!

Types of research question

For what different types of research question might I use a regression model?

Inference / association : Is there a relationship between X & Y ?

• what does the relationship look like?

• Typically want to account for other explanatory variables

• if we have one particular variable of interest, parsimony may be less important (want to control for confounders)

• Model diagnostics { assumptions are important

Other ways of comparing models

Alternatives to AIC: General idea: deviance + penalty

$$\text{AIC: penalty} = 2p$$

$$\text{BIC: penalty} = p \log(n) \quad \leftarrow \text{more stringent}$$

$\Rightarrow$ BIC tends to choose smaller models

LRT: (only nested models)

$$\Lambda = \frac{L_{\text{reduced}}}{L_{\text{full}}}$$

$$-2 \log \Lambda \approx \chi^2_{\epsilon} \quad (\text{under } H_0)$$

ϵ = difference in # parameters

$$-2 \log \Lambda = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

$$\text{reject when} \quad > \chi^2_{\epsilon, \alpha}$$

Activity

Work on the class activity (handout).

Activity

LRT:

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

Reject when $G > \chi^2_{q,\alpha}$.

Activity

- ▶ LRT will choose the full model if $G > \chi^2_{q,\alpha}$
- ▶ AIC will choose the full model if $G > 2q$
- ▶ LRT is more strict when $\chi^2_{q,\alpha} > 2q$

```
alpha = 0.05
```

```
q <- 1:10
```

```
qchisq(1 - alpha, q) - 2*q
```

```
## [1] 1.84145882 1.99146455 1.81472790 1.48772904 1.10128473
```

```
## [7] 0.06714045 -0.49268694 -1.08102240 -1.69296195 -2.28312988
```

```
alpha = 0.01
```

```
q <- 1:20
```

```
qchisq(1 - alpha, q) - 2*q
```

```
## [1] 4.634897e+00 5.210340e+00 5.344867e+00 5.276704e+00
```

```
## [6] 4.811894e+00 4.475307e+00 4.090235e+00 3.665994e+00
```

```
## [11] 2.724970e+00 2.216967e+00 1.688250e+00 1.141238e+00
```

```
## [16] -7.309118e-05 -5.913364e-01 -1.194694e+00 -1.809131e+00
```

AIC vs. LRT

- ▶ LRT can only be used with nested models; AIC can compare non-nested models
- ▶ LRT is more stringent when q (difference in number of parameters between models) is small
- ▶ AIC is more stringent when q is large
- ▶ But, they ask different questions!
 - ▶ LRT: Is difference in model deviance due to chance?
 - ▶ AIC: Estimate expected prediction error
- ▶ **Neither** is asking whether difference in models is **meaningful** (*practical significance*)